

Continuous Glucose Monitoring

ACG: A-0126 (AC)
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- Clinical Indications
- Alternatives
- Evidence Summary
 - Background
 - Criteria
 - Inconclusive or Non-Supportive Evidence
 - Rationale
 - Related CMS Coverage Guidance
- References
- Footnotes
- Codes

Clinical Indications

- Continuous glucose monitoring may be indicated when **ALL** of the following are present(1)(2)(3)(4)(5)(6)(7)(8):
 - Diabetes (eg, type 1 diabetes mellitus, type 2 diabetes mellitus, insulin-deficient diabetes)
 - Type of monitoring is **1 or more** of the following:
 - Long-term continuous glucose monitoring^[A] needed, as indicated by **1 or more** of the following(38)(39)(40):^[N]
 - Adult or child treated with intensive insulin regimen (eg, 3 or more daily insulin injections, use of continuous subcutaneous insulin infusion pump)(46)(58)(59)(60)
 - Adult or child with type 1 diabetes
 - Adult treated with basal-only insulin, with or without noninsulin antihyperglycemic therapy(61)
 - Short-term continuous glucose monitoring needed, as indicated by **ALL** of the following(62):^[N]
 - Additional information about blood glucose needed, as indicated by **1 or more** of the following:
 - Assessment of glycemic control (eg, time in target range) needed for treatment decision making
 - Dawn phenomenon,^[B] known or suspected
 - Hypoglycemic unawareness (ie, patient does not have symptoms with hypoglycemia)^[C](25)(63)
 - Nocturnal hypoglycemia, known or suspected
 - Postprandial hyperglycemia, known or suspected
 - Significant change to diabetes treatment regimen (eg, initiation of insulin, change from multiple-dose insulin to insulin pump therapy)
 - Unexplained hyperglycemia
 - Monitoring limited to 3 to 14 days(69)(70)(71)

Alternatives

- Alternatives include self-monitoring of blood glucose with home glucose monitor.(1)(25)

Evidence Summary

Background

Continuous glucose monitoring is accomplished by use of a small catheter inserted into subcutaneous tissue.(9)(10)(11) **(EG 2)** Electrochemical sensors currently available commercially for measuring subcutaneous glucose levels are able to provide an adequate glucose signal for 7 to 14 days.(9)(10)(11)(12)(13) **(EG 2)** Continuous glucose monitoring devices are designed for either long-term use (also called "personal" monitors) or short-term use (also called "professional" or "retrospective diagnostic" monitors).(9)(12)(14)(15)(16) **(EG 2)** Long-term monitors are usually owned by the patient and display glucose levels in real time or near real time, allowing the patient to adjust their insulin dose or food intake. Short-term monitors are usually provided by a clinician and allow for collection of glucose levels over a period of a few days; glucose levels may be available to the patient during use, or not displayed to the patient but available for the clinician's review at the end of use.(9)(12)(14)(15) **(EG 2)** Long-term continuous glucose monitors can be real-time systems, which transmit a continuous stream of data to the user in real time with alarms for abnormal glucose values, or intermittently scanned systems, which continuously collect data but require the user to actively transmit sensor data to a reader; some intermittently scanned continuous glucose monitors have the capacity to generate alarms for abnormal glucose values.(17)(18) **(EG 2)**

The accuracy of continuous glucose monitoring devices is often expressed as the mean absolute relative difference (MARD), which is the average of the differences in glucose measurements between the device and a reference standard (typically serum glucose), reported as a percent of the reference value.(19)(20)(21)(22) **(EG 2)** A MARD of 10% or less has been proposed as an acceptable limit for safe use of continuous glucose monitoring devices.(19)(23)(24) **(EG 2)** MARD values may be influenced by study design, making inter-study comparisons of MARD values problematic.(21)(22)(23) **(EG 2)**

The patient or the patient's caregiver should receive education and training regarding the proper use of the continuous glucose monitor before implementation.(11)(25)(26)(27)(28) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 55 published peer reviewed articles, 12 specialty society or other evidence-based guidelines, 1 Cochrane systematic review, and 2 book sections.

For diabetes, and need for long-term continuous glucose monitoring, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A systematic review and a clinical practice guideline recommend continuous glucose monitoring in patients with type 1 and type 2 diabetes mellitus, particularly for those who use the device frequently.(41)(42) **(EG 1)** A meta-analysis including 21 randomized controlled trials (2252 patients) evaluated the effectiveness of continuous glucose monitoring compared with self-monitoring of blood glucose on glucose control in patients with type 1 diabetes and found that continuous glucose monitoring was associated with a reduction in HbA1c and a lower risk of severe hypoglycemia (ie, requiring hospitalization or help from a third party) compared with self-monitoring of blood glucose.(43) **(EG 1)** A multicenter randomized controlled trial compared continuous glucose monitoring with self-monitored blood glucose in 203 patients age 60 years and older with type 1 diabetes mellitus. At 6-month follow-up, a small but significant reduction in time spent in hypoglycemia (ie, glucose less than 70 mg/dL (3.89 mmol/L)) was observed in the intervention arm only, most notably in patients with worse baseline levels of hypoglycemia. Ten patients in the control arm experienced severe hypoglycemia requiring assistance as compared with one patient in the continuous glucose monitoring arm.(44) **(EG 1)** An industry-sponsored, multicenter, randomized controlled trial compared continuous glucose monitoring with self-monitored blood glucose in 153 type 1 diabetes mellitus patients age 14 to 24 years. Significant reduction in mean HbA1c occurred in the intervention arm only; adherence with the continuous glucose monitor at 26-week follow-up was 68%.(45) **(EG 1)** An extension study of a randomized controlled trial enrolled 107 patients with type 1 diabetes treated with multiple daily injections and found, at a median follow-up of 1.2 years, that continuous glucose monitoring was associated with a sustained decrease in HbA1c and time spent in hypoglycemia, and an increase in overall well-being (as measured by the World Health Organization 5-item well-being index) and treatment satisfaction (as measured by the Diabetes Treatment Satisfaction Questionnaire). The authors noted that the extension study was limited by the lack of a parallel control group and loss of participants from the original randomized controlled trial.(46) **(EG 2)** A multicenter open-label randomized controlled trial of 156 patients age 16 years and older with type 1 diabetes mellitus and elevated HbA1c (all of whom were treated at baseline with continuous insulin infusion or multiple daily injections) compared intermittently scanned continuous glucose monitoring including optional alarms for low blood glucose (FreeStyle Libre 2) with self-monitored blood glucose and found, after 24 weeks, that continuous glucose monitoring was associated with a lower HbA1c and a higher proportion of time spent in the target glucose range compared with self-monitored blood glucose.(47) **(EG 1)** A multicenter, open-label, randomized controlled trial evaluated continuous glucose monitoring vs self-monitored blood glucose alone in 215 pregnant patients with type 1 diabetes mellitus on intensive insulin therapy and reported a small yet significant reduction in HbA1c, less time in hyperglycemia, and more time in target glucose range in the continuous glucose monitoring arm. Both arms experienced similar hypoglycemia event rates and time in hypoglycemia. Neonatal secondary outcomes included fewer large-for-gestational-age babies and reduced number of stays in neonatal intensive care lasting longer than 24 hours in the continuous glucose monitoring arm.(48) **(EG 1)** A retrospective review of 396 children, adolescents, and adults with newly diagnosed type 1 diabetes found that initiation of continuous glucose monitoring within 1 year of diagnosis was associated with better glucose control and fewer diabetes-related emergency visits, irrespective of therapy with insulin pump or multiple daily injections.(49) **(EG 2)** A consensus specialty society guideline recommends using continuous glucose monitoring in young children who have significant variability in glucose levels or difficulty in identifying hypoglycemic episodes.(50) **(EG 2)** A national guideline recommends that continuous glucose monitoring, along with an educational program to support its use, be offered to all children with type 1 diabetes, and to children with type 2 diabetes who are unable to engage in self-monitoring of blood glucose levels, would otherwise be advised to self-monitor at least 8 times per day, have recurrent or severe hypoglycemia, or are on insulin therapy.(42) **(EG 2)** A systematic review and meta-analysis of 26 randomized controlled trials (2783 patients) comparing at least 8 weeks of intervention with 2 or more glucose monitoring systems (real-time continuous glucose monitoring, intermittently scanned continuous glucose monitoring, or self-monitoring of blood glucose) in patients with type 2 diabetes found that continuous glucose monitoring (either real-

time or intermittently scanned) was associated with greater improvement in HbA1c compared with self-monitoring of blood glucose. The authors noted that the analysis was limited by small sample sizes and heterogeneity in populations, interventions, and outcome definitions.(51) **(EG 1)** A subgroup analysis of a pragmatic randomized controlled trial included patients with both type 1 (34 patients) and type 2 (82 patients) diabetes mellitus age 60 years and older and compared continuous glucose monitoring with self-monitoring of blood glucose; greater improvement in HbA1c was found in patients in the continuous glucose monitoring arm. No severe hypoglycemia or diabetic ketoacidosis occurred in either arm.(52) **(EG 1)** A national specialty society guideline recommends the use of real-time or intermittently scanned continuous glucose monitors for adult and pediatric patients with type 1 diabetes or with type 2 diabetes who are treated with multiple daily insulin injections or continuous subcutaneous insulin infusion therapy, and for adults with type 2 diabetes who are treated with basal-only insulin.(1) **(EG 2)** Expert consensus statements and clinical practice guidelines recommend that continuous glucose monitoring be considered for all patients with type 1 diabetes mellitus, as well as patients with type 2 diabetes mellitus who are receiving intensive insulin therapy, in order to improve glycemic control and reduce hypoglycemia.(2)(4)(38)(53)(54) **(EG 2)** Expert consensus guidelines and a review article suggest that continuous glucose monitoring be considered for patients with cystic fibrosis-related diabetes and diabetes in the setting of chronic pancreatitis.(55)(56)(57) **(EG 2)**

For diabetes, and need for short-term continuous glucose monitoring, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A systematic review of 3 studies (157 participants) on the use of continuous glucose monitoring in adults with type 1 diabetes and impaired awareness of hypoglycemia concluded that, after patients complete a structured educational program, continuous glucose monitoring can reduce rates of severe hypoglycemia and improve hypoglycemia awareness, without worsening glycemic control or restoring counter-regulatory hormone responses.(63) **(EG 1)** Short-term continuous glucose monitoring may help tailor therapy in patients with type 2 diabetes mellitus who are experiencing hypoglycemia or hypoglycemia unawareness.(64) **(EG 2)** Expert consensus guidelines suggest that blood glucose measures derived from continuous glucose monitoring, such as time in target range and time above target range, be used in conjunction with HbA1c as a treatment target and to inform changes in therapy.(6)(7)(8)(65) **(EG 2)** A national specialty society guideline recommends assessment of glycemic control with HbA1c, continuous glucose monitoring, or both at least twice a year in patients with diabetes.(1) **(EG 2)** An expert consensus guideline recommends the use of short-term continuous glucose monitoring in patients with newly diagnosed type 2 diabetes mellitus or unexplained hypoglycemia.(66) **(EG 2)**

Inconclusive or Non-Supportive Evidence

For implantable sensors for continuous glucose monitoring, evidence is insufficient, conflicting, or poor and demonstrates an incomplete assessment of net benefit vs harm; additional research is recommended. **(RG B)** A prospective, single-arm, unblinded, multicenter study evaluated the accuracy of an implantable glucose sensor in 181 adults with type 1 or type 2 diabetes, 96 of whom received a second sensor in their other arm; for the second sensor, 53 patients received a second primary sensor and 43 patients received a modified next-generation sensor. Over 180 days of follow-up, the mean absolute relative difference relative to clinic-based blood glucose measurements (over a blood glucose range of 40 to 400 mg/dL (2.22 to 22.20 mmol/L)) was 10.5% for the primary sensor and 8.5% for the modified sensor; 90% of the modified sensors survived.(29) **(EG 2)** A 90-day, single-arm, blinded, prospective study evaluated the accuracy of an implantable glucose sensor in 90 adults with type 1 or type 2 diabetes; the mean absolute relative difference was 8.8% over a blood glucose range of 40 to 400 mg/dL (2.22 to 22.20 mmol/L), and 91% of sensors functioned through 90 days. The trial was limited by underrepresentation of non-White patients and by the lack of data on long-term effects, including the long-term impact of replacing the sensing device every 3 months.(30) **(EG 2)** In a multicenter prospective open-label trial of 35 adults with type 1 or type 2 diabetes, the mean absolute relative difference of an implantable glucose sensor was 9.6% over 90 days for a blood glucose reference range of 40 to 400 mg/dL (2.22 to 22.20 mmol/L); all sensors remained functional through the 90-day trial. The authors noted that further studies are needed to assess the accuracy and safety of serial insertions and removals of the device.(31) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(72); 42 CFR 422.101(73)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(74); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(75); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(76)

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Footnotes

[A] Self-monitoring of blood glucose is required to calibrate some continuous glucose monitoring systems(32)(33)(34) and to assist in making acute treatment decisions when using continuous glucose monitoring devices approved only for adjunctive use. Self-monitored blood glucose measurement is also appropriate whenever a concern arises about the accuracy of a continuous glucose monitor reading.(1)(20)(34)(35)(36)(37) [A in Context Link 1]

[B] The dawn phenomenon is an abnormal elevation of blood glucose in the early morning hours.(67) [B in Context Link 1]

[C] Hypoglycemia unawareness is caused by a patient's inability to release counter-regulatory hormones in response to hypoglycemia. This commonly occurs in patients with long-standing type 1 diabetes mellitus, as a result of autonomic neuropathy, or in very young children.(1)(67)(68) [C in Context Link 1]

Codes

CPT®: 0446T, 0447T, 0448T, 95249, 95250, 95251

HCPCS: A4238, A4239, A9276, A9277, A9278, E2102, E2103, G0564, G0565, S1030, S1031

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